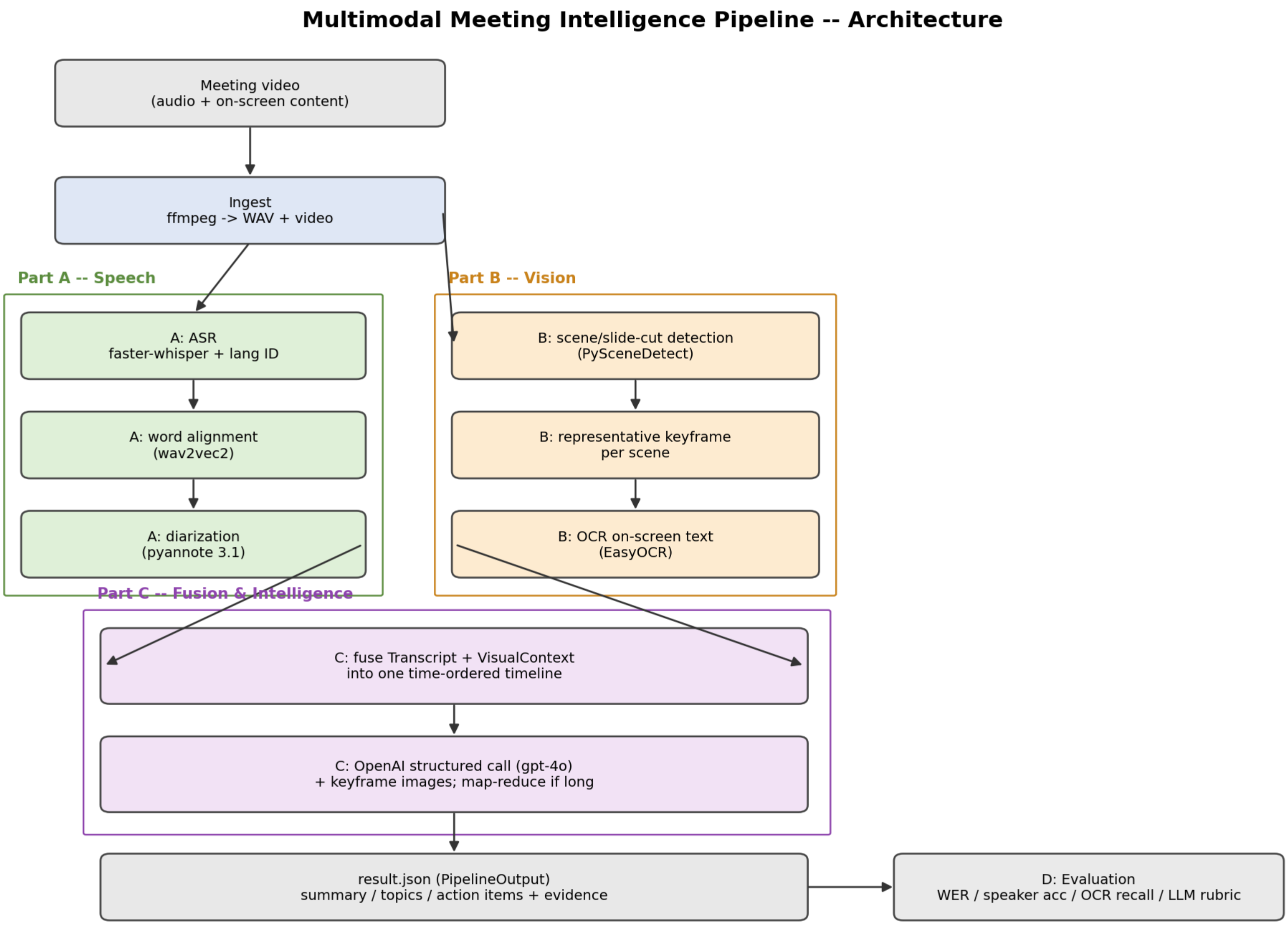
Multimodal Meeting Intelligence Pipeline — Executive Report

A 26-minute council-meeting recording in, a grounded, evidence-linked meeting summary out. This report covers the architecture, key decisions, evaluation results, failure modes, and resource footprint.

1. Architecture

The pipeline is a linear cascade of independent, swappable stages connected by explicit Pydantic data contracts ([`src/mmi/schemas.py`]): speech and vision run in parallel on the same source video, fuse onto one time-ordered timeline, and are handed to an LLM that must ground every claim in timestamped evidence.



Design choices that matter:

- **WhisperX + pyannote 3.1** for ASR/diarization — VAD-batched inference and word-level alignment give materially better speaker attribution than a naive faster-whisper + clustering glue, at the cost of a heavier dependency stack.
- **Scene-cut detection + one keyframe/scene (PySceneDetect + EasyOCR)** instead of processing every frame — captures slide content at a fraction of the cost.
- **Vision-capable LLM captioning** — rather than building a dedicated face/person-detection stage, representative keyframe images are attached directly to the `gpt-4o` call. The model reads slide text and layout from the image itself, which is more robust than OCR alone (see §3).
- **Strict schema output** (OpenAI structured outputs / Pydantic) instead of free-form text, so every summary/topic/action item carries `Evidence` (timestamp, speaker, scene, quote) and the result is machine-consumable.
- **Map-reduce chunking** above a configurable character budget, so meeting length doesn't blow the context window.

- **Disk-cached stage artifacts** (`outputs/cache/*.json`) so an expensive ASR pass isn't repeated while iterating on prompts or re-running only the LLM stage.

2. Model / API choices

Stage	Choice	Why	Trade-off
ASR	WhisperX (`small`)	Fast CTranslate2 backend, auto language ID, word alignment	`small` trades WER for fitting a 4 GB GPU; `medium`/`large-v3` configurable
Diarization	pyannote 3.1	State-of-the-art open diarizer, wired into WhisperX	Needs an HF token; sensitive to overlapping speech
Scene detection	PySceneDetect (content)	Cheap, robust to slide/scene cuts	Threshold tuning; gradual fades can be missed
OCR	EasyOCR (GPU)	Simple install, GPU-accelerated	Weaker on dense/stylised slide text (swappable via config)
LLM	OpenAI `gpt-4o` (vision)	Native multimodal input, strong structured-output support	Higher cost/latency than `gpt-4o-mini`; hosted dependency

3. Evaluation summary

Metrics are computed by `evaluate.py` against a small hand-labelled reference. Full numbers and methodology are in `report.md`; headline results on the provided clip:

Stage	Metric	Result
ASR	Word Error Rate (first 60 s window)	6.67 %
Diarization	Speaker attribution accuracy (first 60 s)	100 % (2 speakers in window; 6 globally)
Vision/OCR	Keyword recall across keyframes	100 % (1/1 term); 15/16 scenes yielded text
Intelligence	Grounded-evidence fraction	100 % (9/9 items: 7 topics, 1 action item, 1 decision)

Failure analysis:

- *Language ID on noisy intros* — Whisper's majority-vote language detector mis-identified the (English) recording as Maori (`mi`) on all three sampled windows, likely triggered by genuine Maori place names ("Waikato", "Waipa District"). The transcript stage stores `language="mi"`, but the LLM stage self-corrects to `"en"` using full-meeting context. A `--language en` CLI override is available but not forced by default.
- *OCR misses/garbles small or stylised slide text* — e.g. "€" substituted for "&", partial word recognition on a dense process-diagram slide. This is the main reason vision-capable LLM captioning was added: the LLM reads the same keyframe image and recovers content OCR could not (verified — see `scene_id 12`'s "Mitigate / Restart / Reimagine" slide in `outputs/result.json`).
- *LLM stage is a single external dependency* — an OpenAI quota/network error is caught and the run still emits a valid partial `result.json` (`intelligence=null`, `metadata.llm_error` populated) rather than failing the whole pipeline; the (expensive) transcript/visual artifacts are not lost. The committed `result.json` is a clean run where the LLM call succeeded (`intelligence` populated, `llm_error: null`).
- *Diarization on overlapping speech / short backchannels* remains the largest general risk to speaker-accuracy on longer, less orderly meetings.

Latency & resource notes (this run: 26 min / 1560 s clip, 4 GB T1200 laptop GPU, `small` Whisper, `gpt-4o` with keyframe images; ASR/alignment/diarization/ LLM timings from a `--language en` run, vision timing from a separate deterministic scene-detect + OCR pass on the same clip/machine — vision output doesn't depend on transcript language, so the two are directly comparable):

Stage	Wall-clock	Notes
Ingest (ffmpeg)	~2 s	Cached WAV extraction
Speech — ASR	~161 s	faster-whisper `small`, float16, batch 8
Speech — alignment	~38 s	wav2vec2, skipped if no model for detected language
Speech — diarization	~117 s	pyannote 3.1, CPU/GPU depending on availability
Vision — scene detect + OCR	~63 s	PySceneDetect + EasyOCR, 16 scenes
Intelligence — LLM call	~24 s	`gpt-4o`, 16 keyframe images attached, map-reduce (2 chunks)
Total	**~407 s (~6.8 min)**	End-to-end for a 26 min / 1560 s meeting

GPU memory is the binding constraint on the reference machine (4 GB), so the speech stage loads/uses/frees each model (ASR → align → diarize) sequentially rather than holding them all resident. Vision and the LLM call are CPU/network bound and comparatively cheap. Cost is dominated by LLM tokens and image count (capped via `max_keyframe_images`, default 40, evenly sampled for longer meetings) — see [`config.yaml`](config.yaml).

4. Production considerations

- **Scaling**: stages are stateless given their inputs → each maps to its own horizontally-scaled worker/queue; long meetings are handled via VAD-chunked ASR, scene-bounded (duration-independent) CV, and LLM map-reduce.
- **Monitoring**: cache per-stage artifacts (already done) and run the eval harness on a labelled golden set in CI to track WER, speaker-accuracy, OCR recall, and grounded-fraction drift over time; log detected language, speaker count, and scene count per run as cheap health signals.
- **Robustness**: missing HF token → diarization skipped (single speaker) instead of crashing; no CUDA → CPU fallback; OCR/alignment failures are caught per-item so one bad frame/segment can't fail the run; LLM failures degrade to a partial artifact instead of data loss.
- **With one more week**: active-speaker detection (face + lip-motion) to fuse *who is on screen* with diarization; a real DER via `pyannote.metrics`; retrieval-grounded citations; a golden-set CI job.